```
CREATE DATABASE course10;
USE course10;

DESC movie_table;

SET SQL_SAFE_UPDATES=0;

UPDATE movie_table
SET release_date = STR_TO_DATE(release_date, '%d-%m-%Y');
ALTER TABLE movie_table
MODIFY release_date DATE;
```

#How has movie release changed over the years

```
SELECT YEAR(release_date) AS release_year,
COUNT(title) AS movie_count
FROM movie_table
GROUP BY YEAR(release_date)
ORDER BY release_year;
```

release_year	movie_count
1916	1
1925	1
1927	1
1929	2
1930	1
1932	1
1933	2
1934	1
1935	1
1936	2
1937	2
1938	2
1939	3
1940	5
1941	1
1942	2
1944	2
1945	4
1946	3
1947	3
1948	3
1949	2
1950	1
1951	3
1952	4
1953	4
1954	4
1955	2
1956	3
1957	2
1958	1
1959	3
1960	3
1961	5
1962	7
1963	8
1964	10
1965	8
1966	6
1967	4
1968	10
1969	10
1970	12
1971	10
1972	9
1973	9
1974	8
1975	6
1976	9
1977	16

1978	14
1979	16
1980	22
1981	33
1982	24
1983	23
1984	28
1985	28
1986	26
1987	31
1988	31
1989	32
1990	29
1991	30
1992	34
1993	47
1994	55
1995	70
1996	97
1997	112
1998	133
1999	171
2000	166
2001	183
2002	203

2003	169
2004	204
2005	217
2006	238
2007	195
2008	227
2009	247
2010	225
2011	223
2012	208
2013	231
2014	238
2015	216
2016	104
2017	1

#How has popularity changed over the years from 1970

```
SELECT YEAR(release_date) AS release_year,
ROUND(AVG(popularity), 2) AS avg_popularity,
COUNT(title) AS movie_count
FROM movie_table
WHERE YEAR(release_date) >= 1970
GROUP BY YEAR(release_date)
ORDER BY release_year;
```

release_year	avg_popularity	movie_count
1970	7.05	12
1971	11.92	10
1972	12.01	9
1973	21.16	9
1974	19.9	8
1975	19.75	6
1976	16.49	9
1977	14.74	16
1978	13.07	14
1979	15.82	16
1980	15.61	22
1981	10.51	33
1982	18.45	24
1983	15.16	23
1984	18.66	28
1985	15.12	28
1986	16.61	26
1987	17.45	31
1988	16.65	31
1989	16.4	32
1990	24.53	29
1991	14.69	30
1992	19.61	34
1993	15.58	47
1994	16.08	55

1995	17.2	70
1996	13.73	97
1997	15.61	112
1998	13.98	133
1999	12.25	171
2000	14.11	166
2001	14.25	183
2002	14.02	203
2003	14.84	169
2004	17.38	204
2005	14.79	217
2006	13.93	238
2007	16.68	195
2008	17.07	227
2009	15.99	247
2010	17.26	225
2011	19.28	223
2012	19.09	208
2013	20.65	231
2014	16.31	238
2015	16.51	216
2016	23.64	104
2017	0.71	1

#How has the Profit of movies significantly changed before and after the year 2000

```
SELECT CASE WHEN YEAR(release_date) < 2000 THEN 'Before 2000'
ELSE 'After 2000' END AS period,
ROUND(AVG(revenue - budget), 2) AS avg_profit,
ROUND(SUM(revenue - budget), 2) AS total_profit,
COUNT(title) AS movie_count
FROM movie_table GROUP BY period;
```

period	avg_profit	total_profit	movie_count
After 2000	17606169.52	61533562485	3495
Before 2000	17366754.54	22715714935	1308

 # How revenue changed over the year

```
SELECT YEAR(release_date) AS release_year,
ROUND(AVG(revenue), 2) AS avg_revenue,
COUNT(title) AS movie_count
FROM movie_table
GROUP BY YEAR(release_date)
ORDER BY release_year;
```

release_year	avg_revenue	movie_count
1916	8394751	1
1925	22000000	1
1927	650422	1
1929	2179000	2
1930	8000000	1
1932	25	1
1933	2240500	2
1934	4500000	1
1935	3202000	1
1936	5618000	2
1937	92462743	2
1938	5716550.5	2
1939	20841656	3
1940	36558400	5
1941	6000000	1
1942	14816250.5	2
1944	6127500	2
1945	6783000	4
1946	17898041.33	3
1947	2600000	3
1948	3989333.33	3
1949	6600000	2
1950	8000000	1
1951	16333333.33	3
1952	12800000	4

	release_year	avg_revenue	movie_count
	1953	20000000	4
	1954	13267960.25	4
	1955	3500000	2
	1956	11266666.67	3
	1957	17150000	2
	1958	17570324	1
	1959	16666666.67	3
	1960	22466666.67	3
	1961	14891364.4	5
	1962	30010747.29	7
	1963	35730404.5	8
	1964	34701446	10
	1965	35964028.5	8
	1966	9891666.67	6
	1967	31948084.5	4
	1968	18755447.7	10
	1969	31438585.4	10
	1970	20668065.75	12
	1971	29624179.8	10
	1972	7276254.56	9
	1973	59717054.22	9
	1974	48270896.75	8
	1975	33835595.17	6
	1976	33148207.78	9
	1977	26207017.5	16

	1978	48462056.71	14
	1979	61859555.69	16
	1980	35079656.91	22
	1981	28611290.79	33
	1982	44937422.54	24
	1983	44344423.65	23
	1984	40810275.14	28
	1985	30559112.61	28
	1986	32892264.65	26
	1987	31469691.68	31
	1988	42550695.48	31
	1989	40942836.22	32
	1990	47696266.9	29
	1991	41889861.43	30
	1992	51556031.32	34
	1993	30214821.55	47
	1994	48934617.45	55
	1995	44793619.53	70
	1996	38985844.45	97
	1997	45756552.86	112
	1998	33796539.47	133
	1999	29460095.21	171
	2000	40578488.21	166
	2001	36074699.34	183
	2002	38040725.66	203
	2003	43272872.59	169
	2004	43038432.32	204
	2005	40192885.49	217
	2006	36892310.26	238
	2007	37202288.58	195
	2008	41043166.72	227
	2009	31535257.6	247
	2010	36182331.64	225
	2011	46337465.69	223
	2012	36356126.59	208
	2013	37164363.72	231
	2014	33247752.05	238
	2015	33350594.13	216
	2016	46682894.7	104
	2017	0	1

#Has the popularity of movies significantly changed before and after the year 2000

```
SELECT CASE WHEN YEAR(release_date) < 2000 THEN 'Before 2000' ELSE 'After 2000' END AS period,
ROUND(AVG(popularity), 2) AS avg_popularity
FROM movie_table
WHERE release_date IS NOT NULL
GROUP BY period;
```

period	avg_popularity
After 2000	16.68
Before 2000	15.02

#Compare average popularity and average rating.

```
SELECT CASE WHEN YEAR(release_date) < 2000 THEN 'Before 2000' ELSE 'After 2000' END AS period,
   ROUND(AVG(popularity),2) AS avg_popularity,
   ROUND(AVG(vote_average),2) AS avg_rating
FROM movie_table
WHERE release_date IS NOT NULL
GROUP BY period;
```

period	avg_popularity	avg_rating
After 2000	16.68	6.15
Before 2000	15.02	6.44

 # Stored procedure for getting movie details using title

```
CALL fetch_details ('Avatar');

CREATE PROCEDURE `fetch_details` (in name varchar(200))
BEGIN
SELECT * FROM movie_table WHERE title=name;
END
```

budget	genres	id	keywords	original_language	overview	popularity
15000000.0	Action,Adventure,Fantasy,Science Fiction	19995.0	culture clash,future,space war,space colony,so...	en	In the 22nd century, a paraplegic Marine is disp...	12.921594

production_companies	production_countries	release_date	revenue	runtime	status	tagline	title	vote_average
Ingenious Film Partners,Twentieth Century Fox ...	United States of America,United Kingdom	2009-12-10	19170001.0	103.0	Released	Enter the World of Pandora.	Avatar	7.2

vote_count
235.0